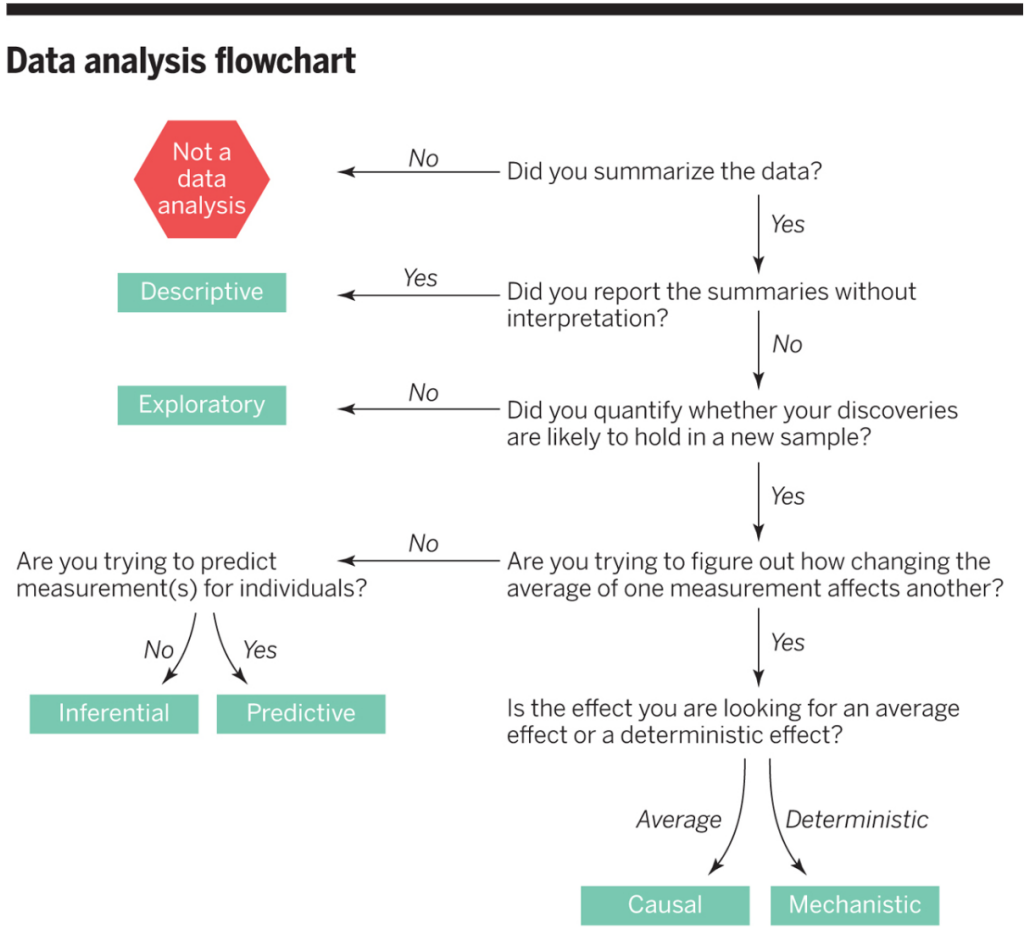
What is the question?

Practice for DSCI 522 data analysis project

Materials created by Tiffany Timbers & Gabriela Cohen Freue

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What is the question?



What is the question? by Roger Peng and Jeff Leek

So you know the type of question, now what?

This helps narrow down the possibilities of the kind of analysis you might want to do!

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For example, if you have the question: **"How many viral illnesses will someone have next year?"** and you identify that it is **predictive.** You could narrow down that some kind of statistical or machine learning model might help you answer that.

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Then you need to go a step deeper and look at the data that you have, and see which kind of statistical or machine learning model is most suitable for your data.

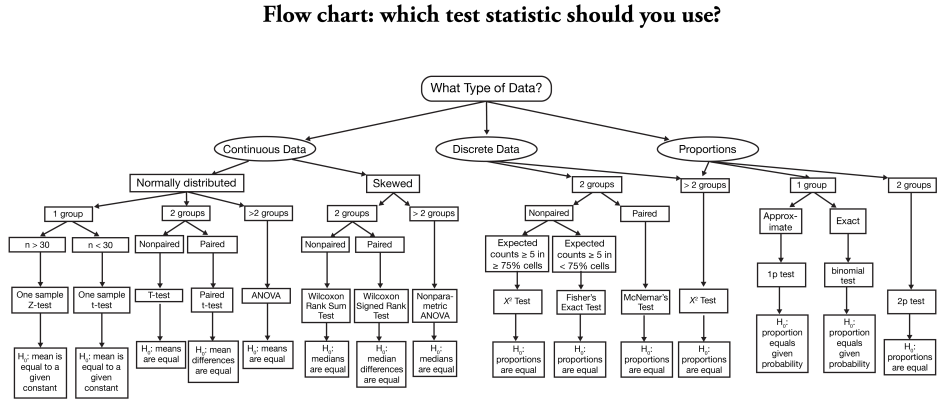
Source: scikit-learn algorithm cheat sheet

Another example

For example, if you have the question: "Is the gestational length of first born babies the same as that of non-first borns?" and you identify that it is inferential. You could narrow down that some kind of statistical inference approach might help you answer that.

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Then again, you need to go a step deeper and look at the data that you have, and see which kind of statistical inference approach is most suitable for your data.



Source: https://onishlab.colostate.edu/summer-statistics-workshop-2019/

Or for another, see this website: http://www.biostathandbook.com/testchoice.html

Practice

Case 1

Question: *Is a yet undiagnosed patient's breast cancer tumor malignant or benign?*

Data:

ID	Radius	Texture	Perimeter	Area	Smoothness	Class
926125	1.9275296	1.3485941	2.1001278	1.9667039	0.9627130	M
926424	2.1091388	0.7208383	2.0589739	2.3417954	1.0409262	M
926682	1.7033556	2.0833009	1.6145108	1.7223261	0.1023682	M
926954	0.7016669	2.0437755	0.6720844	0.5774446	-0.8397450	M
927241	1.8367249	2.3344032	1.9807813	1.7336925	1.5244257	M
92751	-1.8068114	1.2207179	-1.8127934	-1.3466044	-3.1093489	B

Data reference: https://archive-beta.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+diagnostic

Case 2

Question: *Is inhalation of marijuana associated with lung cancer?*

Data:

ID	sex	gender	age	smoker	marijuana_use	bmi	lung_cancer
57749	male	fluid	35	1	never	22.3	0
54903	male	male	43	0	frequent	18.0	0
55430	male	male	29	0	sometimes	32.5	1
52626	female	female	54	0	frequent	20.0	0
56638	female	female	37	0	never	26.1	0
51969	male	male	51	0	never	29.8	1

Note: this is simulated data.

Case 3

Question: *Does a truncation of the BRCA2 gene cause cancer?*

Data:

ID	sex	gender	age	smoker	bmi	brca2_truncation	cancer
21654	male	fluid	35	1	22.3	1	1
21966	male	male	43	0	18.0	0	1
23029	male	male	29	0	32.5	1	1
29838	female	female	54	0	20.0	0	0
24964	female	female	37	0	26.1	0	0
20907	male	male	51	0	29.8	0	0

Note: this is simulated data.

Some key notes:

- identifying whether there even is a response variable is important!
- the kind of response variable/target is critical for narrowing down the method
- the explanatory variables/predictors/features are also important, but I consider these after the response variable

How to approach your analysis for this course

1. Identify the kind of question
2. Look at the data
3. Identify a **suitable** statistical method for your question and data
4. Create a visualization
5. Apply your statistical method
6. (maybe create another visualization)
7. Interpret and communicate your assumptions and results (in the context of the question you originally asked)

Questions?